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Compressive sensing sectional imaging for single-shot in-line self-interference incoherent holography

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ABSTRACT

A numerical reconstruction method based on compressive sensing (CS) for self-interference incoherent digital holography (SIDH) is proposed to achieve sectional imaging by single-shot in-line self-interference incoherent hologram. The sensing operator is built up based on the physical mechanism of SIDH according to CS theory, and a recovery algorithm is employed for image restoration. Numerical simulation and experimental studies employing LEDs as discrete point-sources and resolution targets as extended sources are performed to demonstrate the feasibility and validity of the method. The intensity distribution and the axial resolution along the propagation direction of SIDH by angular spectrum method (ASM) and by CS are discussed. The analysis result shows that compared to ASM the reconstruction by CS can improve the axial resolution of SIDH, and achieve sectional imaging. The proposed method may be useful to 3D analysis of dynamic systems.

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1. Introduction

Digital holography [1–3] enables direct access to the reconstruction of the object wavefront. However, coherent illumination can introduce disadvantageous effects into the holographic recording and reconstructing process, such as speckle noise and parasitic interference. Therefore, incoherent digital holography is becoming increasingly attractive for the elimination of these sources of noise [4–7] and has been applied to adaptive optics [8,9], fluorescence microscopy [10,11], ambient light color holography [12] and so on. Usually, the phase-shifting method is employed for the elimination of the zero order and twin image terms, and angular spectrum method (ASM) is applied for the image reconstruction. However, there may be some difficulties for dynamic applications when employing phase-shifting methods which require multiple interferogram recording per complex hologram. Furthermore, by ASM, the in-focus reconstructed image would be disturbed by the out-of-focus section information. These issues limit the application of SIDH in sectional imaging. Compressive sensing (CS) [13,14], a rapidly growing signal acquisition paradigm, has been successfully applied to coherent digital holography for three-dimensional (3D) tomography via in-line holography [15–17], for the recovery of partially occluded object [18,19], and in incoherent digital holography for 3D scene recording by multiple view projection [20]. Sectional imaging based on optical scanning holography is achieved by using inverse imaging [21,22].

Here we propose a numerical algorithm for achieving tomographic imaging by applying CS to SIDH reconstruction. Numerical simulation and experiments demonstrate the feasibility and validity of the method. The intensity distribution along the propagation direction and the axial resolution of SIDH by ASM and by CS are discussed. The axial resolution of SIDH by ASM is typically worse than that of a classical imaging system [23]. However, our analysis shows that the CS method can narrow the axial width of the image spot size and improve the axial resolution of SIDH compared to ASM. Furthermore, it can achieve sectional imaging by significantly suppressing the signals from out-of-focus sections of the object volume.

2. Theory

2.1. Basic theory of SIDH

A basic SIDH apparatus is illustrated in Fig. 1. The two curved mirrors (M_A and M_B) with different focal lengths (f_A and f_B) are equidistant from the beam splitter (BS). The BS produces two copies of the spherical wave emanating from each point-source of a diffuse object, and recombines the reflections from the two mirrors with different curvatures. A Fresnel zone-like interference pattern results from each source point of the object. The intensity

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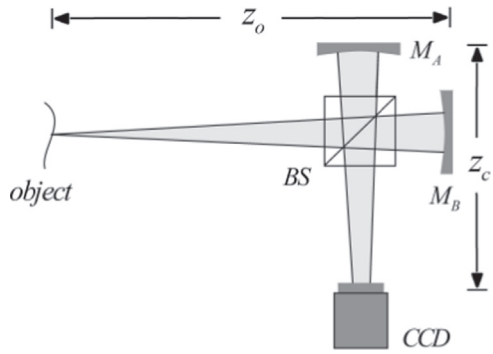

Fig. 1. Basic configuration of SIDH. BS, beam splitter; M_A, curved mirror of focal length f_A; M_B, curved mirror of focal length f_B.

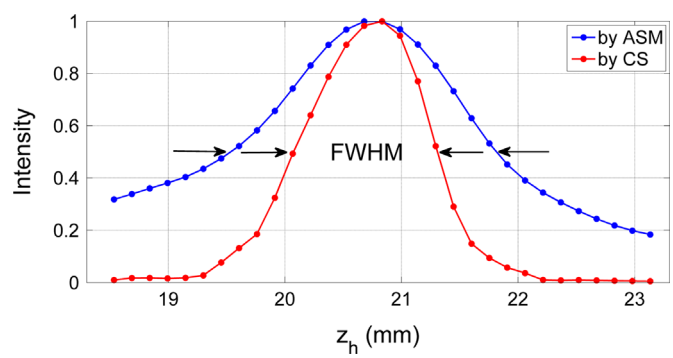

Fig. 2. Intensity distribution by ASM and by CS along z direction, where point-source of $\lambda = 625$ nm at $z_o = 150$ mm and hologram of size 256×256 pixels (pixel pitch 6 μm × 6 μm) recorded at $z_c = 175$ mm with $f_A = 100$ mm and $f_B = 75$ mm.

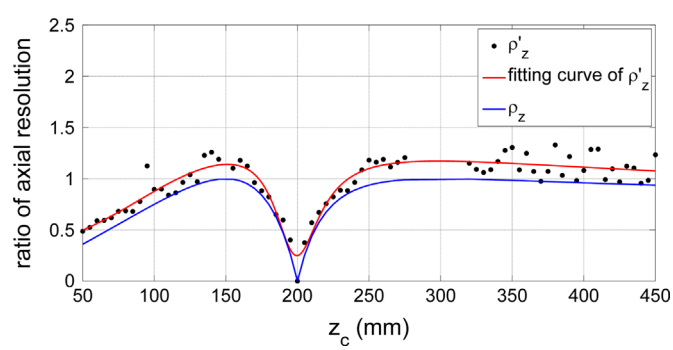

Fig. 3. Axial resolution ratio by ASM ρ_z and by CS ρ'_z vs. z_c, where point-source of $\lambda = 625$ nm at $z_o = 150$ mm and hologram at $z_c \in (50, 400)$ mm with $f_A = 100$ mm, $f_B = 75$ mm and $R = 3$ mm. (For interpretation of the references to color in this figure, the reader is referred to the web version of this article.)

pattern at the CCD plane is an incoherent superposition over all the point-sources of the object as:

$$h_c(x_c, y_c) = \iint dx_o dy_o I_o(x_o, y_o)\left\{\left[|E_A(x_c, y_c)|^2 + |E_B(x_c, y_c)|^2\right]\right\} + I'_o(x_c, y_c) \odot Q_{z_h}(x_c, y_c) + I'_o(x_c, y_c) \odot Q_{-z_h}(x_c, y_c) \quad (1)$$

where the symbol "$\odot$" represents the convolution; I_o is the intensity distribution of the object; $I'_o(x_c, y_c) = I_o(x_c/\alpha, y_c/\alpha)$ with $\alpha = -z_c/z_o$; E_A and E_B are the optical fields arriving at CCD after reflection from the two mirrors; λ is the wavelength; $Q_{z_h}(x_c, y_c) = \exp[ik(x_c^2 + y_c^2)/2z_h]$; and z_h is the holographic image distance given by:

$$z_h(z_o) = -\frac{\left[\frac{z_o f_A}{z_o - f_A} - z_c\right]\left[\frac{z_o f_B}{z_o - f_B} - z_c\right]}{\frac{z_o f_B}{z_o - f_B} - \frac{z_o f_A}{z_o - f_A}} \quad (2)$$

For holographic reconstruction, the single-shot in-line self-interference incoherent hologram is back propagated by ASM with the distance $-z_h$ under Fresnel or paraxial approximation as:

$$I = h_c \odot Q_{-z_h} \quad (3)$$

However, the expression (1) includes disturbance of the zero-order and the twin image terms. The reconstructed image also contains mixture of in-focus and out-of-focus contributions.

2.2. Axial resolution of SIDH

The axial resolution of SIDH is discussed in comparison with the classical imaging system. Firstly, according to Eq. (2), we obtain the axial magnification of SIDH as:

$$M_z = \left|\frac{dz_h}{dz_o}\right| = \left|-\frac{f_{AB} z_c^2}{z_o^2}\left(\frac{1}{\zeta_A} + \frac{1}{\zeta_B}\right)\right| \quad (4)$$

where $\frac{1}{f_{AB}} = \frac{1}{f_A} - \frac{1}{f_B}$, $\frac{1}{\zeta_A} = \frac{1}{z_o} + \frac{1}{z_c} - \frac{1}{f_A}$, and $\frac{1}{\zeta_B} = \frac{1}{z_o} + \frac{1}{z_c} - \frac{1}{f_B}$. The axial resolution of SIDH in the reconstructed image space δ_{z_h} is proportional to $\lambda z_h^2/R_H^2$. The radius of the hologram, $R_H = |R(z_\kappa - z_c)/z_\kappa|$, is given by the smaller of the two projections of the mirrors on the CCD plane, where R is the radius of the mirror and $z_\kappa = z_o f_\kappa/(z_o - f_\kappa)$, with $\kappa = A$ or B. The axial resolution SIDH in the object plane is defined as:

$$\delta_{z_o} = \left|\frac{\delta_{z_h}}{M_z}\right| \quad (5)$$

For the classical imaging system, the axial resolution in the object space $\overline{\delta_{z_o}}$ is proportional to $\lambda z_o^2/R^2$, and the ratio of the axial resolution of the classical system to that of SIDH is:

$$\rho_z = \left|\frac{\overline{\delta_{z_o}}}{\delta_{z_o}}\right| \quad (6)$$

The larger the values of ρ_z the better the resolution of the SIDH system. However, ρ_z of SIDH is between 0 and 1, i.e. it means that the axial resolution of SIDH would be worse than that of the classical imaging system [23]. So, it is disadvantage for achieving sectional imaging.

2.3. Algorithm of SIDH by CS

In order to achieve sectional imaging without the disturbance from zero-order, twin, and out-of-focus components, we apply CS to SIDH reconstruction. Firstly, the nonlinearity caused by the zero-order term $|E_A|^2 + |E_B|^2$ is regarded as model error e, the twin image term $I'_o \odot Q_{-z_h}$ and the other out-of-focus information is considered as n. Therefore, for 3D scene recording, Eq. (1) is rewritten as:

$$h_c = I'_o \odot Q_{z_h} + n + e = \Im_{2D}^{-1}\left\{\int \Im_{2D}\left[I'_o(x_c, y_c, z_o)\right]\Im_{2D}\left[Q_{z_h(z_o)}(x_c, y_c)\right]dz_o\right\} + n + e \quad (7)$$

where

$$\Im_{2D}\left[Q_{z_h(z_o)}(x_c, y_c)\right] = i\lambda z_h(z_o)\exp\left[-i\pi\lambda z_h(z_o)(\xi^2 + \eta^2)\right] \quad (8)$$

Here $\Im_{2D}\{\}$ and $\Im_{2D}^{-1}\{\}$ represent the two-dimensional (2D) Fourier transform and the inverse 2D Fourier transform; (ξ, η) are frequency coordinates. The discretized hologram of $N_x \times N_y$ pixels with pixel pitch $\Delta_x \times \Delta_y$ is converted into 1D vector as

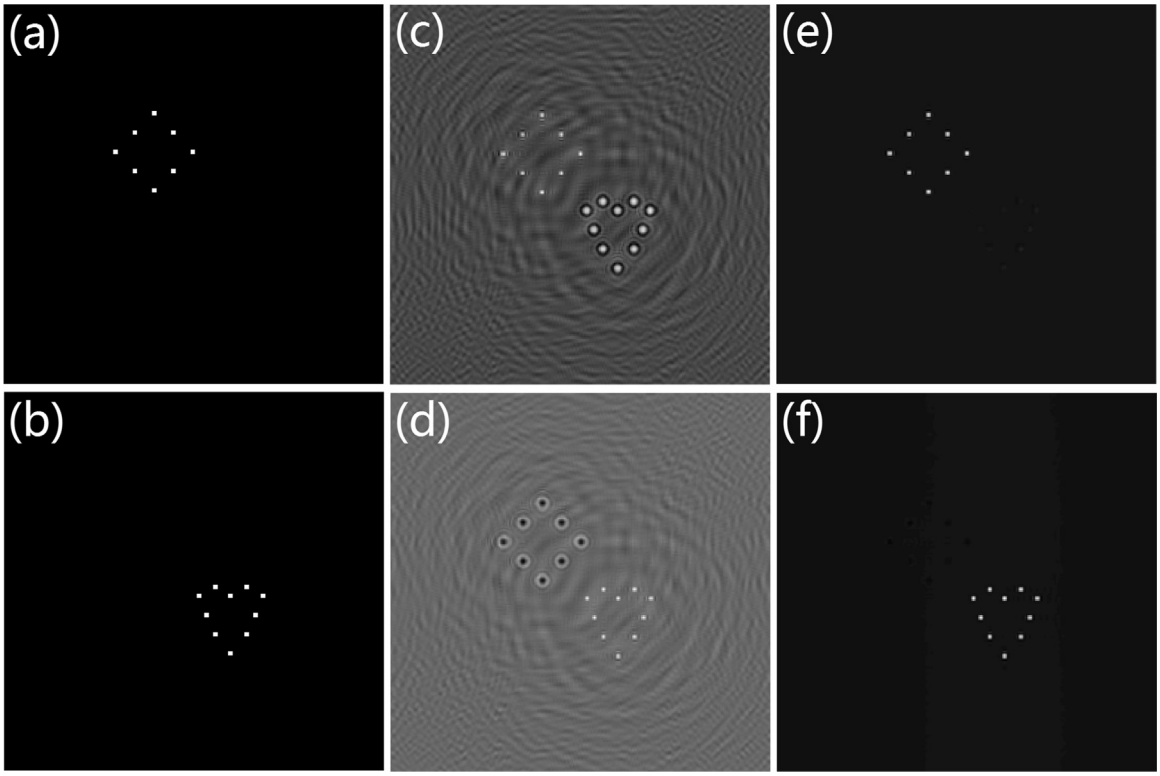

Fig. 4. Numerical simulation. (a)–(b) two target object patterns "diamond" and "heart" at $z_o = 150 \pm 0.8$ mm; reconstructed image (c)~(d) by ASM; (e)~(f) by CS, where single-shot in-line hologram of size 256×256 pixels (pixel pitch $6\ \mu m \times 6\ \mu m$) at $z_c = 175$ mm with $f_A = 100$ mm and $f_B = 75$ mm is employed for reconstruction.

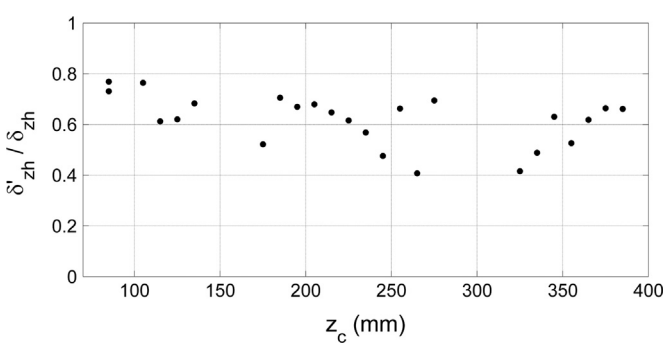

Fig. 5. Ratio $\delta'_{z_h}/\delta_{z_h}$ vs. z_c, where LED of $\lambda = 625$ nm at $z_o = 147$ mm and holograms of size 512×512 pixels (pixel pitch $4.65\ \mu m \times 4.65\ \mu m$) at different z_c with $f_A = 100$ mm, $f_B = 75$ mm.

$g_{(q-1)N_x+p} = h_c(p\Delta_x, q\Delta_y)$. The discretized $I'_o(x_c, y_c, z_o)$ of sampling points $N_x \times N_y \times N_z$ with sampling pitch $\Delta_x \times \Delta_y \times \Delta_z$ is converted into 1D vector as $f_{(l-1)(N_x \times N_y)+(n-1)N_x+m} = I'_o(m\Delta_x, n\Delta_y, l\Delta_z)$. According to the CS process, the SIDH recording expressed by Eq. (7) can be rewritten as:

$$g = \left[F_{2D}^{-1}TB\right]f + n + e = Hf + n + e \tag{9}$$

$B = bldiag(F_{2D}, F_{2D}, \ldots, F_{2D})$ is a block diagonal matrix of size $(N_x \times N_y \times N_z) \times (N_x \times N_y \times N_z)$ with F_{2D} representing the 2D discrete Fourier transform matrix of size $(N_x \times N_y) \times (N_x \times N_y)$. Bf performs the corresponding calculation of $\Im_{2D}\{I'_o(x_c, y_c, z_o)\}$. Define $T = [S_1 S_2 \ldots S_{N_z}]$ with $[S_l]_{m,n} = i\lambda z_h(l\Delta_z)\exp[-i\pi\lambda z_h(l\Delta_z)(m^2\Delta\xi^2 + n^2\Delta\eta^2)]$, where $\Delta\xi = 1/(N_x\Delta_x)$, $\Delta\eta = 1/(N_y\Delta_y)$. F_{2D}^{-1} represents the inverse 2D discrete Fourier transform matrix. The matrix $H = F_{2D}^{-1}TB$ is the sensing operator, which denotes the SIDH recording process. The transformation model expressed by Eq. (9) is inverted by de-compressive inference by enforcing the two-step iterative shrinkage-thresholding algorithm (TwIST) [24] to solve the optimization problem:

$$\hat{f} = \arg\min_f \frac{1}{2}\|g - Hf\|^2 + \tau\|f\|_{TV} \tag{10}$$

where $\hat{f}$ is the estimated result by enforcing TwIST, $\|\cdot\|$ denotes l_2-norm, τ is a regularization parameter, and $\|\cdot\|_{TV}$ denotes the total variation (TV).

We show below that, the axial width of the image spot size in the image space can be narrowed by introducing CS in the reconstruction of SIDH image. Furthermore, the CS decompressive reconstruction method can effectively remove the out-of-focus information. Therefore, compared to ASM backprojection using in-line single-shot interferogram, it can provide increased axial resolution and achieve sectional imaging by CS.

3. Numerical simulation and results

3.1. Improvement of axial resolution

Numerical simulation is performed to characterize ρ_z in comparison between ASM and CS methods. Firstly, a point-source of $\lambda = 625$ nm at $z_o = 150$ mm is employed. The hologram of size 256×256 pixels (pixel pitch $6\ \mu m \times 6\ \mu m$) is recorded at $z_c = 175$ mm with $f_A = 100$ mm and $f_B = 75$ mm. The Point Spread Function (PSF) is an important parameter for the discus of the resolution of the optical image system. The most practical way to determine the resolution of an optical system is to measure the Full-Width at Half-Maximum (FWHM) of the PSF. The larger the FWHM of the PSF, the worse of the resolution. So, for the investigation of the axial resolution of SIDH, the axial FWHM of the 3D PSF, as shown in Fig. 2, is discussed. Fig. 2 shows the intensity of the reconstructed point-source image versus reconstruction distance z_h by ASM and by CS. It shows that the FWHM of the PSF

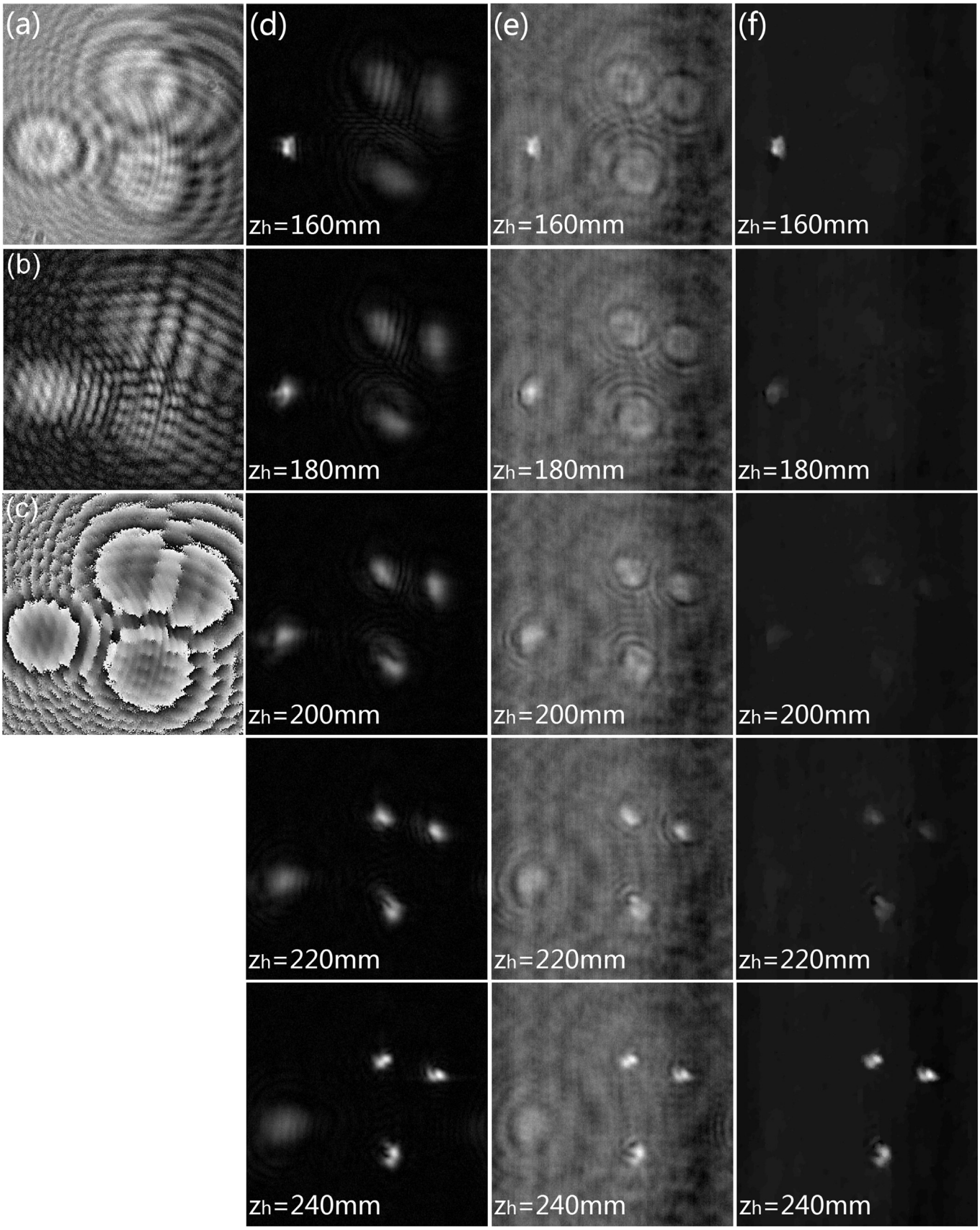

Fig. 6. Experiment of LEDs. (a) one of the interferograms of size 256 × 256 pixels (6.3 mm × 4.8 mm sensor area with 640 × 480 pixels); (b) intensity and (c) phase of the complex hologram by phase-shifting method; reconstructed image (d) by phase-shifting method; (e) by ASM; (f) by CS at different z_h.

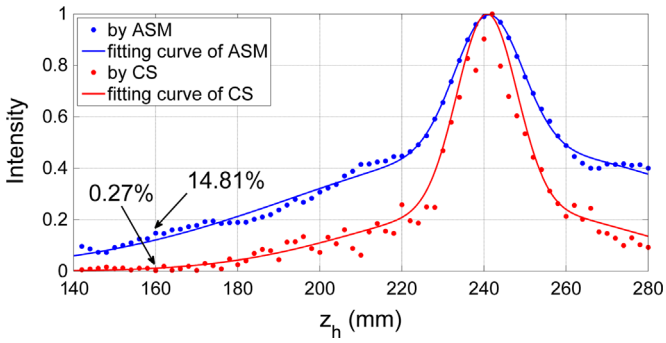

Fig. 7. Intensity LED image, focusing at $z_h = 240$ mm, along z direction by ASM using the complex hologram, and by CS using in-line single-shot interferogram.

by CS is narrower than that by ASM. That is because the PSF depends not only on several experimental parameters but also on the image processing algorithm used to perform the 3D reconstruction. The decompressive inference method in the paper, by enforcing the TwIST algorithm with a nonquadratic TV regularizer which constraints for objects with few edge, aims to directly reconstruct the 3D volume object information rather than the field at a distance which is a superposition of all the object planes each of which is at a different focus. When it is not at the exact z_h position where the object point is, the energy of the reconstructed image would get dispersive, i.e. the edge of the point would get blurred. So, the intensity of the reconstructed point would drop more quickly along axial direction with the increase of the distance away from the exact z_h position by performing the decompressive inference method with TV regularizer. As a result, the FWHM of the PSF by CS would be narrower than that by ASM. It means that the axial resolution in the image space by CS δ'_{z_h}, is smaller than that by ASM δ_{z_h}. That is, $\delta'_{z_h} < \delta_{z_h}$ and ρ_z is improved by CS compared with that by ASM.

In Fig. 3, graphs of $\rho_z(z_c)$ are obtained by calculating a series of holograms recorded at different camera positions z_c and using ASM or CS for reconstruction of images. Here, point-source of $\lambda = 625$ nm is at $z_o = 150$ mm, and hologram is at $z_c \in (50, 400)$ mm with $f_A = 100$ mm, $f_B = 75$ mm and $R = 3$ mm. The values of axial resolution δ'_{z_h} and δ_{z_h} are determined by the full-width-half-maximum of graphs such as in Fig. 2. The black dots in Fig. 3 are $\rho'_z = \left|\overline{\delta_{z_o}}/\delta'_{z_o}\right|$ obtained using CS method. According to the mathematical relationship of z_c and ρ_z, $\rho_z(z_c) = \frac{f_{AB}(z_A - z_B)^2}{\zeta_{AB} z_k^2} \cdot \frac{(z_C - 2\zeta_{AB})(z_K - z_C)^2 z_k^2}{z_C(z_A - z_C)^2(z_B - z_C)^2}$ with $\frac{1}{\zeta_{AB}} = \frac{1}{f_A} + \frac{1}{f_B} - \frac{2}{z_o}$, we can obtain that the numerator and the denominator of the expression are the 5th degree polynomial of z_c. So, $f(x) = \frac{p_1x^5 + p_2x^4 + p_3x^3 + p_4x^2 + p_5x + p_6}{x^5 + q_1x^4 + q_2x^3 + q_3x^2 + q_4x + q_5}$ is employed for curving fitting, as the red curve shown in Fig. 3, where $\begin{cases} p_1 = -696.8, \ p_2 = 5373, p_3 = -2193, p_4 = 301.6, p_5 = -23.4, p_6 = 1.799 \\ q_1 = 6001, \ q_2 = 3459, q_3 = 809.5, \ q_4 = -99.37, q_5 = 5.631 \end{cases}$. The blue line is $\rho_z = \left|\overline{\delta_{z_o}}/\delta_{z_o}\right|$ obtained using ASM. The analysis shows increase of ρ_z and improvement of the axial resolution by introduction of CS in reconstruction of SIDH image.

3.2. Sectional imaging of SIDH by CS

Another numerical simulation with two groups of discrete point-sources of $\lambda = 625$ nm at $z_o = 150 \pm 0.8$ mm, i.e. $\Delta z_o = 1.6$ mm, is performed for the description of sectional imaging. The hologram of size 256×256 pixels (pixel pitch $6\ \mu m \times 6\ \mu m$) is recorded at $z_c = 175$ mm with $f_A = 100$ mm and $f_B = 75$ mm. $\delta_{z_o} = [4\lambda z_h^2/M_z(N_x\Delta_x)^2] = 0.8$ mm corresponding to the axial resolution in the object space. Fig. 4(a) and (b) are the two target patterns "diamond" and "heart" used as the objects at the two distances. It should be noted that, single-shot SIDH theory is employed for analysis here. Fig. 4(c) and (d) shows the reconstructed images of the two patterns calculated by ASM at the corresponding distances. Different sectional images are recognizable for $\delta_{z_o} < \Delta z_o$, but the reconstructed in-focus sectional image is

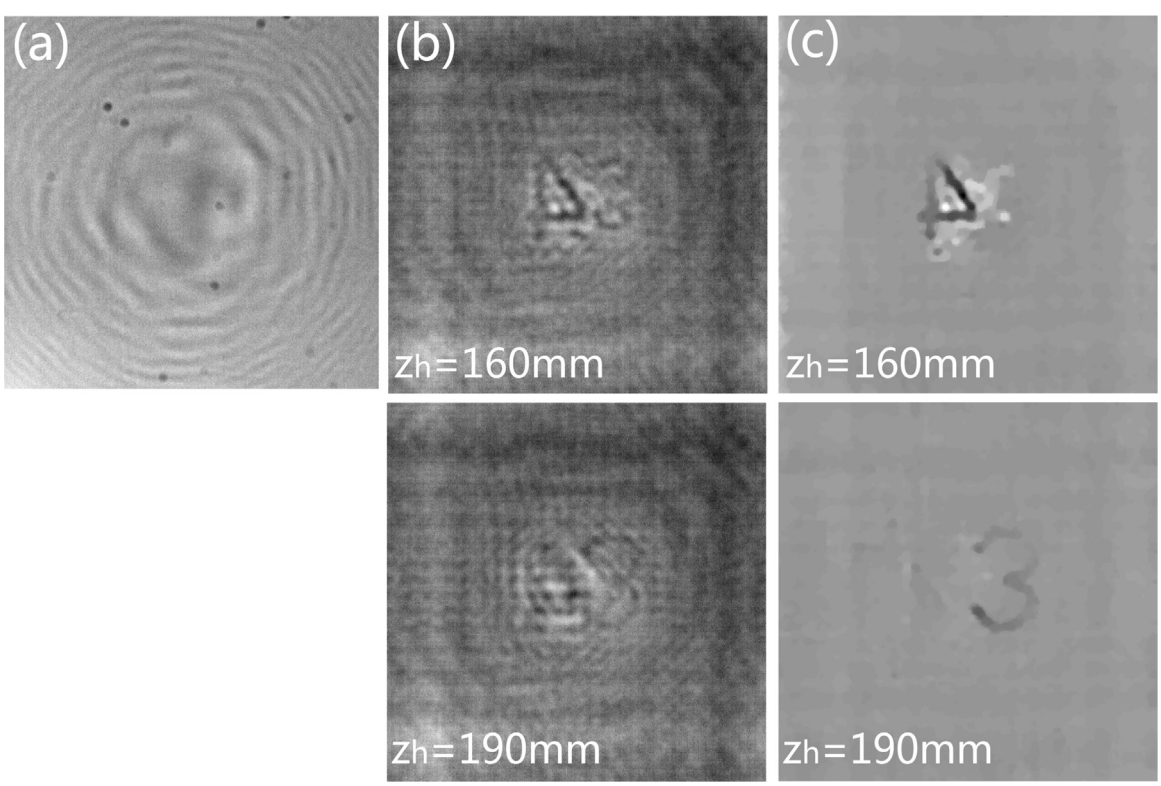

Fig. 8. Experiment of USAF. (a) in-line single-shot interferogram of size 512×512 pixels with pixel pitch $4.65\ \mu m \times 4.65\ \mu m$; reconstructed image (b) by ASM; (c) by CS at $z_h = 160$ mm and 190 mm.

disturbed by the out-of-focus contribution. Fig. 4(e) and (f) shows the reconstructed image by CS. It shows that not only the zero-order and the twin image terms, but also the out-of-focus contribution are well suppressed and true sectional imaging is obtained.

4. Experimental results

4.1. Improvement of axial resolution

Experiments employing LEDs and extended objects are performed to demonstrate the feasibility and validity of the proposed method. Firstly, similar to the numerical simulation of a point-source, an LED ($\lambda = 625$ nm), spatial filtered with a microscope objective (20 ×, 0.4 NA) and a pinhole (50 μm), is set at about $z_o = 147$ mm. The apparatus depicted in Fig. 1 is employed. Holograms of size 512 × 512 pixels (pixel pitch 4.65 μm × 4.65 μm) are recorded at various distances z_c. From each hologram, axial resolution of images reconstructed using ASM and CS methods are determined by the FWHM of the PSF. The ratio of the axial FWHM of the PSF by CS and ASM is plotted vs. z_c in Fig. 5. The ratio remains less than unity throughout the range of z_c, demonstrating the advantage of CS method over ASM in axial resolution.

4.2. Sectional imaging of SIDH by CS

To demonstrate sectional imaging, an experiment uses two groups of LEDs ($\lambda = 625$ nm) placed at different position by a different configuration [9]. Fig. 6(a) displays an interferogram captured by the camera. Fig. 6(b) and (c) is the amplitude and phase of the complex hologram obtained from a combination of four interferograms by the phase-shifting method. The column (d) of Fig. 6 shows images reconstructed by ASM at various distances, starting from the complex hologram. This column of images shows the two groups of LEDs coming in and out of focus at different distances, as expected. Columns (e) and (f) of Fig. 6 show the images reconstructed at corresponding distances, using ASM and CS methods respectively, but starting from a single-shot interferogram. The column (e) represents ASM reconstruction from in-line single-shot interferogram, and contains disturbance from zero-order and twin-image terms. In contrast, CS reconstruction in column (f) effectively suppresses not only the zero-order and twin-image terms but also contribution from out-of-focus sections. Two in-focus sectional images are clearly seen at $z_h = 160$ mm and 240 mm. Fig. 7 shows the axial intensity distribution of one of LED's image that focuses at $z_h = 240$ mm reconstructed by ASM using the complex hologram, and by CS using in-line single-shot interferogram. At $z_h = 160$ mm, the intensity drops to 14.81% by ASM, and to 0.27% by CS. So, the CS method can achieve sectional imaging by significantly suppressing the signals from out-of-focus sections of the object volume.

Finally, two resolution targets (USAF 1951, group 0 numbers “4” and “3”) with diffusers are used as extended objects and located at $z_o = 142$ mm and 130 mm and illuminated by an LED ($\lambda = 625$ nm). Hologram of size 512 × 512 pixels (pixel pitch 4.65 μm × 4.65 μm) is recorded at $z_c = 68$ mm. Fig. 8(a) is an in-line single-shot interferogram. Columns (b) and (C) show the reconstructed images by ASM and by CS, respectively, at two in-focus positions of $z_h = 160$ mm and 190 mm. We find that the CS method effectively suppresses out-of-focus component and better achieves sectional imaging compared to ASM.

5. Conclusion

In summary, a numerical reconstruction of SIDH based on CS method is proposed to achieve sectional imaging for single-shot in-line self-interference incoherent holography. The numerical simulation and experimental analysis demonstrate the feasibility and validity of the proposed method. Firstly, it is seen that the axial FWHM of the PSF can be narrowed by CS. Furthermore, the CS decompressive reconstruction method can suppress not only the zero-order and the twin-image terms, but also the out-of-focus disturbance. Therefore, compared to ASM backprojection, it can provide increased axial resolution and achieve sectional imaging by CS for SIDH. In addition, the proposed method is applicable to single-shot in-line interferogram, instead of multi-frame phase-shifting complex hologram, offering potential advantages in tomographic analysis of dynamic 3D systems.

Acknowledgments

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